

The power of Metabase is its accessibility. Fellow team members across departments can schedule reports, create notifications based on alerts, and share live dashboards without asking engineers for help.

While it may not have the deep analytics or customisation features of heavier BI platforms, Metabase is perfect for organisations that want to move fast, answer everyday questions, and reduce dashboard development friction.

Grafana for metrics and monitoring

Grafana has become the *de facto* platform for visualising metrics, logs, and time-series data in real-time. Grafana was originally a darling of the DevOps and infrastructure monitoring world, but it has grown so popular that it's now being used in many different industries for diverse tasks such as monitoring server health, visualising environmental sensor data, and much more.

This robust platform integrates with a multitude of data sources ranging from Prometheus to InfluxDB, Elasticsearch, PostgreSQL, and even Google Sheets! Users can create highly customisable dashboards visualising real-time graphs/alerts that can spike or channel a series of events based on specified thresholds. Its array of dynamic querying options and templated dashboards make Grafana an excellent use case for large-scale monitoring solutions.

Grafana is not limited to visualisation; it is powerful in alerting and observability too. Grafana supports multi-channel alerts to streamline how you respond to incidents. There are also several plugins available to add to your experience with unique visualisations, data sources and authentication methods.

Overall, what really makes Grafana stand out is its extensibility and flexibility. It can act as a centralised source for figuring out complex systems, and it keeps things straightforward so that operational

data is easily accessible for everyone including engineers and executives.

Plotly and Dash: Building interactive web-based visuals

Plotly and Dash are a powerful combination for developing interactive, web-based data applications. Plotly is the plotting library that can make high-quality plots—everything from simple line and bar charts to complex 3D graphs and choropleth maps. It's flexible and works not only in Python but also in R and JavaScript too. It is widely appreciated for its aesthetics and interactivity.

Dash is based on Plotly and gives developers the ability to build full-fledged data applications in Python. Setting up controls such as buttons, sliders, and dropdowns, and linking them to your visualisations is incredibly easy! It allows quick prototyping of dashboards, modelling tools, and applications in the analytics space. What distinguishes Dash from notebooks or static reports is that it is easy to interact with even if a user doesn't have a technical background. Furthermore, Dash supports authentication, theming, and access to deployment options such as Flask and Docker.

Polars + Jupyter: High-speed DataFrames meet visualisation

As datasets keep growing, pandas can reach limits. Enter Polars, a lightning-fast DataFrame library, built in Rust and made for heavy lifting of data. If you pair it with Jupyter, Polars can preserve that exploratory experience, while enhancing performance when it counts (10x faster), especially on multi-core machines.

Polars emphasises lazy evaluation (you won't execute anything until you are ready), memory efficiency, and parallelism by default. It can be used for very large CSVs, time series data, or streaming logs that would either overwhelm or crash pandas (and Python).

Although Polars is very fast, it uses a syntax that feels familiar to pandas users, which also helps learn Polars. When analysts use it with Jupyter, they can issue rapid queries, create visualisations in Plotly or seaborn, and iterate quickly on very large data.

In summary, Polars provides a Jupyter experience with the best of both worlds -- a fast backend, and an effective and interactive caching frontend for analysis and story-telling. It is quickly becoming an analyst and data engineer's choice when performance is the priority.

Hands-on projects: A few examples

The following hands-on projects mix storytelling with technology. These examples illustrate how open source tools can transform raw data into engaging narratives.

Visualising IPL stats using Python and Matplotlib

Cricket is a sport that lends itself easily to data, especially when we talk about the IPL (Indian Premier League). This data project with Python, pandas and Matplotlib can help explore player performance, team wins, and matches using datasets from the IPL. First, we need to clean the IPL CSV data and then use groupby functions and pivot tables to summarise and report statistics (for example, total runs, strike rates, and wickets). After the data is summarised, Matplotlib and seaborn can be used to visualise it using colourful bar plots, pie charts and line graphs. You can track Virat Kohli's run totals over seasons, and identify which teams are better at chasing runs. This project helps with data wrangling and visualising data in a sport that has millions of fans.

Real-time IoT sensor data with Grafana and InfluxDB

IoT devices generate a lot of data—typically, time series data for things like temperature, humidity, motion,